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Research on online anomaly detection methods for bearing degradation

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Abstract

In industrial applications, rolling bearings operate under conditions of high precision and high speed, and their physical and mechanical characteristics change with the increase in operating time. Traditional diagnostic methods struggle to adapt well to the changing characteristics of bearings for online anomaly detection. Therefore, this research proposes an online anomaly detection method for rolling bearings based on time-density-weighted incremental support vector data description (TISVDD). A classification strategy is proposed to prevent samples misclassification in the updating process. The Detection Boundary is established based on SVDD decision boundary to enhance the recognition of abnormal samples in the process of model updating. A dual-screening mechanism update strategy for support vectors is proposed. It involves establishing a preliminary screening mechanism based on the Elimination Boundary. On this basis, an in-depth screening mechanism based on time density weight is introduced by considering spatiotemporal characteristics of samples, enhancing the real-time performance of online anomaly detection for bearings. Building upon the fused dual-boundary SVDD, a TISVDD framework for online anomaly detection is proposed, enabling the detection model to dynamically update in response to data changes over time. To validate the effectiveness of the proposed method, experiments were conducted using the XJTU-SY bearing dataset and real-time datasets collected on an online hardware platform. The results demonstrate the effectiveness and superiority of the method in practical applications.

Keywords: anomaly detection, dual-screening, rolling bearing, SVDD, online update

1. Introduction

Bearings, as common components in the industrial field, are also crucial parts of rotating machinery. About 44% of machine failures in the industrial field are related to bearing faults [1]. Therefore, research on bearing anomaly detection is essential. However, the performance of bearings is not constant throughout the entire industrial operation process [2]. With the accumulation of operating time, the performance of bearings gradually degrades. The original model will no longer adapt to the changes in new data, and the model needs to be updated online to follow the changes in bearing performance for real-time and accurate judgment of the bearing's status. Hence, the research significance of online bearing anomaly detection is significant.

Currently, online anomaly detection for bearings mainly focuses on the vibration signals of the bearings. Research methods are mainly divided into those based on physical characteristics and data-driven methods [3, 4]. In physical characteristics-based methods, the study primarily involves researching online bearing anomaly detection through mathematical modeling and signal processing based on the inherent characteristics of bearings. Jiang and Zhang [5] proposes a method for bearing vibration modeling and online fault size estimation, achieving model-based online fault

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detection and diagnosis. Wang *et al* [6] introduces an online speed-independent order tracking method, enabling online fault diagnosis of bearings under variable speeds. Liang *et al* [7] suggests a combined method of maximum correlation kurtosis deconvolution and band-pass filtering, enabling online detection of early weak structural damage in inter-shaft bearings of aircraft engines. Sharma *et al* [8] proposes establishing a sensitive sub-band selection criterion based on empirical wavelet transform of vibration signals, enabling online fault detection for constant and variable speed bearings. In data-driven methods, most current research on online anomaly detection tends to favor neural networks. Mao *et al* [9] proposes a rule-adaptive neural network architecture, constructing a deep multi-task single-class anomaly detection model. Mao *et al* [10] suggests constructing a deep dual-time domain adaptation model based on transfer learning, achieving early online fault detection of bearings through offline learning of degraded bearing sequences and online processing. Zhou *et al* [11] introduces a time-varying online transfer learning model for intelligent fault diagnosis of rolling bearings, achieving intelligent bearing fault diagnosis with incomplete and unlabeled target data. Liu *et al* [12] proposes integrating domain-adaptive deep stacked denoising autoencoder networks into an online early fault detection framework based on granularity feature prediction, achieving effective feature extraction for online bearing fault data. In the above methods, physically-based approaches generally require prior knowledge of bearings, making it challenging to address the complex variations and unknown fault modes in online bearing anomaly detection. Due to the difficulty in collecting sufficient bearing data in real scenarios, the data in the feature space tends to be imbalanced and unevenly distributed [13, 14]. As a result, in data-driven approaches, methods leaning towards neural networks are prone to overfitting as a result. Additionally, neural network methods rely heavily on a large number of samples and have high computational complexity, making it challenging to improve the real-time performance of online bearing anomaly detection in resource-limited situations.

Support vector data description (SVDD) [15], as another data-driven anomaly detection method, requires fewer samples compared to neural networks and can still perform reliably in anomaly detection with a limited number of samples. Additionally, it has a relatively lower computational complexity, making it easier to train and implement in environments with limited resources. However, traditional SVDD, when used for anomaly detection, remains unchanged once the model is trained. Since the performance of bearings degrades over long-term operation, the SVDD hypersphere model used for anomaly detection needs to follow the dynamic changes in bearing performance. Therefore, incremental SVDD (ISVDD) is more suitable for online bearing anomaly detection. The standard ISVDD [16] updates the model with all new samples each time, although it can demonstrate good performance, it incurs a significant time cost, resulting in a substantial reduction in model update speed. KISVDD [17] and NISVDD [18] train with new samples beyond the original model range and old samples near the decision boundary of the original model. KISVDD selects old samples for updating based on their distribution, while NISVDD selects old samples for updating based on the distribution of new samples. Both methods reduce the number of samples involved in the update to some extent, reducing computation and improving update speed. However, because they only observe the distribution of new or old samples within the original model range, their models exhibit lower robustness. Additionally, the boundary between categorizing samples outside the original model range as either abnormal or degenerated normal samples is unclear, leading to lower accuracy in anomaly detection. HISVDD [19] integrates the accumulated fewer new samples, enhanced through data augmentation, with the old samples near the original model's decision boundary for joint training. This approach enhances the robustness of the model. However, in situations with high sampling frequency, the introduction of data augmentation leads to a decrease in the model's updating speed. Additionally, the anomaly detection rate is relatively low, similar to the methods mentioned above.

Based on the above analysis, three key challenges in ISVDD-based bearing online anomaly detection can be highlighted.

(1) The model faces challenges in defining abnormal samples during the updating process, making it prone to samples misclassification, resulting in false alarms and false positives, ultimately leading to a lower bearing anomaly detection rate.
(2) While maintaining high robustness, the model faces difficulties in improving update speed, as it requires a large number of samples for comprehensive training, resulting in lower real-time capabilities for bearing online anomaly detection.
(3) The model struggles with stable online updates, failing to adapt to the real changes in bearing data, resulting in the difficulty of robust online anomaly detection for bearings.

In response to the above three challenges, through careful analysis, we can draw the following three conclusions:

(1) When bearings degrade in performance, there are normal samples nearby and abnormal samples farther away from the original model. Therefore, the separation of normal and abnormal samples is crucial for improving the bearing anomaly detection rate.
(2) During the updating process, a large number of samples participate in the update training, leading to a decrease in update speed. Therefore, selecting fewer suitable samples for updates is crucial for improving the real-time performance of bearing online anomaly detection.
(3) The model cannot adapt to the changes in online real bearing data, with a large number of mixed normal and abnormal samples participating in training updates. Therefore, establishing an online update scheme that removes abnormal samples and selects appropriate normal samples is crucial for stable online bearing anomaly detection.

Based on the above analysis, this research proposes an online adaptive updating SVDD anomaly detection method. The main contributions of this research are as follows:

(1) A classification strategy is proposed to prevent samples misclassification in the updating process. The Detection Boundary is established based on SVDD decision boundary to improve the model's identification of abnormal samples and to improve the accuracy of bearing online abnormal detection.
(2) A dual-screening mechanism update strategy for support vector is proposed. It involves establishing a preliminary screening mechanism based on the Elimination Boundary. On this basis, an in-depth screening mechanism based on time density weight is established combining with the spatiotemporal characteristics of samples. This approach aims to enhance the model's update speed, consequently improving the real-time capability of bearing online anomaly detection.
(3) A time-density-weighted ISVDD (TISVDD) framework for online anomaly detection is proposed. Based on the fused dual-boundary SVDD, the samples are then given time density weights for screening, so that the model is dynamically updated in time dimension with data changes to achieve robust online anomaly detection of bearings.

The rest of this research is organized as follows: section 2 introduces the necessity of model updates based on ISVDD, and a classification strategy to prevent misclassification of samples during the updating process, a dual-screening mechanism update strategy for support vectors (SVs), and an online anomaly detection framework based on TISVDD are proposed. Section 3 includes a comparative experiment with other methods and presents the experimental results. Lastly, section 4 provides a comprehensive conclusion for the entire research.

2. Proposed method

The online anomaly detection model based on ISVDD does not consider the delineation between anomalous samples and degraded normal samples during model updates, leading to a decrease in anomaly detection accuracy. Simultaneously, the involvement of a large number of new or old samples in updating the model results in a decrease in the model update speed, thereby reducing the real-time capability of bearing online anomaly detection. The existence of these issues can make bearing online anomaly detection challenging to achieve robustly.

In this section, to improve the discrimination accuracy of new samples and simultaneously enhance the speed of model updates for robust online anomaly detection of bearings, this research, starting from the necessity of ISVDD, first proposes a classification strategy to prevent samples misclassification during the updating process. By establishing the Detection Boundary, it can effectively discriminate anomalous samples during the model's online updating. Additionally, a dual-screening mechanism update strategy for SVs is introduced. It establishes the elimination boundary, conducts preliminary screening on samples that may become new support vectors (nSVs), and then further screening through a time-density-weighted update strategy. This reduces the number of samples involved in updates under higher robustness, thereby improving the speed of model updates. Finally, the TISVDD framework for online anomaly detection is proposed. Built on the foundation of a fused dual-boundary SVDD, the samples are then given time density weights for screening. This allows the model to dynamically update in the time dimension, achieving robust online anomaly detection for bearings.

2.1. ISVDD introduction

SVDD aims to identify the SVs of target samples in multidimensional space to provide the optimal data description, thereby constructing the minimum hypersphere, thus improving classification performance. Assuming there exists a dataset $X = \{x_1, x_2, \ldots, x_N\}$, in which N denotes the count of target data points. The problem is described as follows:

$$\begin{aligned} &\min R^2 + C\sum_{i=1}^{n}\xi_i \\ &\text{s.t.} \,||\Phi\,(x_i) - c||^2 \leqslant R^2 + \xi_i, \forall i \end{aligned} \tag{1}$$

where $\xi_i \geqslant 0\,(i = 0, 1, \ldots, n)$, R is the radius of the hypersphere, C is a penalty factor, and ξ_i is the relaxation factor. $\Phi\,(x_i)$ represents the nonlinear mapping function of the sample x_i in the high-dimensional feature space. The volume of the hypersphere is minimized by minimizing R^2. To solve the above optimization problem, constructing the Lagrangian function allows the transformation of the optimization problem into the following dual problem:

$$\begin{aligned} &\min \sum_{i=1}^{n}\alpha_i\,(\Phi\,(x_i)\cdot\Phi\,(x_j)) - \sum_{i=1}^{n}\sum_{j=1}^{n}\alpha_i\alpha_j\,(\Phi\,(x_i)\cdot\Phi\,(x_j)) \\ &\text{s.t.} \sum_{i=1}^{n}\alpha_i = 1, 0 \leqslant \alpha_i \leqslant 1 \end{aligned} \tag{2}$$

where $\alpha_i \geqslant 0$ is the Lagrange multiplier. Solving this optimization problem yields:

$$c_i = \sum_{i=1}^{n}\alpha_i\Phi\,(x_i) \tag{3}$$

$$\begin{aligned} R^2 &= ||\Phi\,(x_{\text{sv}}) - c_i||^2 \\ &= (\Phi\,(x_{\text{sv}})\cdot\Phi\,(x_{\text{sv}})) - 2\sum_{i=1}^{n}\alpha_i\,(\Phi\,(x_i)\cdot\Phi\,(x_{\text{sv}})) \\ &\quad + \sum_{i=1}^{n}\sum_{j=1}^{n}\alpha_i\alpha_j\,(\Phi\,(x_i)\cdot\Phi\,(x_j)). \end{aligned} \tag{4}$$

For an unknown sample z, the distance from z to c is:

$$d^2 = ||\Phi(z) - c_i||^2 = (\Phi(z)\cdot\Phi(z)) - 2\sum_{i=1}^{n}\alpha_i(\Phi(x_i)\cdot\Phi(z)) + \sum_{i=1}^{n}\sum_{j=1}^{n}\alpha_i\alpha_j(\Phi(x_i)\cdot\Phi(x_j)). \quad (5)$$

If $d^2 \leqslant R^2$, then z is within the hypersphere and belongs to the normal samples.

For the ISVDD algorithm, new samples enter the classifier in batches over time for online anomaly detection. When α_i achieves the optimal solution, the observed the sample x_i satisfies the Karush–Kuhn–Tucker (KKT) conditions:

$$\begin{array}{l}\alpha_i = 0 \Rightarrow d^2 < R^2 \\ 0 < \alpha_i < C \Rightarrow d^2 = R^2 \\ \alpha_i = C \Rightarrow d^2 > R^2\end{array}. \quad (6)$$

Samples x_i that satisfy $0 < \alpha_i < C$ are SVs, thereby constructing the hypersphere decision boundary. For newly added samples, if they meet the KKT condition $d^2 < R^2$, then the SVs remains unchanged, and the model does not need to be updated. However, if there are samples among the newly observed samples that violate the KKT conditions (SVKs), the previously computed Lagrange multiplier α_i will no longer satisfy the new model. In this case, the model needs to be updated to meet the requirements of online anomaly detection for bearings [20]. Regarding model updates, the discrimination of online anomalous samples and the improvement of model update speed are important research directions in online anomaly detection.

2.2. A classification strategy to prevent samples misclassification during the updating process

In the context of the ISVDD method, for the discrimination of online anomalous samples, if there are SVKs in the observed samples, typically all SVKs participate in model updates. Subsequently, the updated model is used to perform online anomaly detection on the new samples from the current test set. However, in practice, there may be anomalous samples within SVKs. If all SVKs are included in model updates, anomalous samples may also become SVs, causing potential misclassification of samples during the updating process. Consequently, this can lead to a reduction in the adaptability of the updated model, so that the accuracy of bearing online anomaly detection decreases.

Addressing the challenge of misclassification of anomalous samples in the online bearing anomaly detection mentioned above, we propose a classification strategy to prevent samples misclassification during the updating process, based on the design concept of the dual-boundary SVDD [21] for handling misclassified samples. Detection boundaries are established outside the hypersphere decision boundary, as illustrated in figure 1.

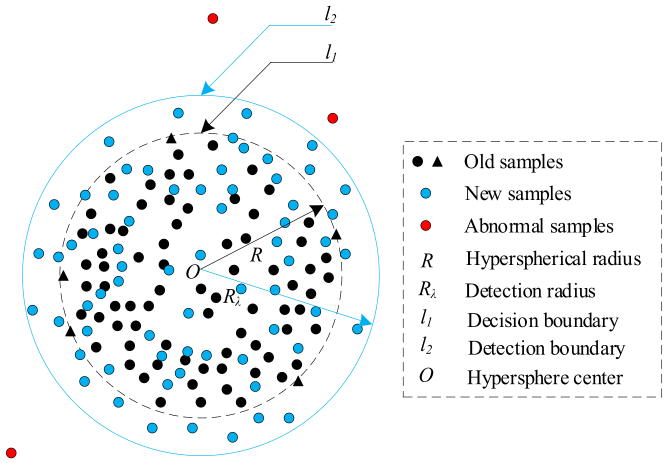

Figure 1. Schematic of the detection boundary.

The establishment of the Detection Boundary outside the hypersphere decision boundary aims to resolve the determination of anomalous samples during the model updating process. By setting an appropriate detection threshold and establishing a suitable Detection Boundary, the objective is to ensure that degraded normal samples during the model updating process fall within the Detection Boundary, while anomalous samples fall outside the detection boundary. Setting the detection threshold:

$$\lambda = \frac{R}{R_\lambda},\ 0 < \lambda \leqslant 1 \quad (7)$$

where R_λ is the radius of the detection boundary. The minimum value of R_λ cannot be less than R. If it is less than R, it may misclassify normal samples violating the KKT conditions as anomalous samples. When equal to R, the detection boundary coincides with the decision boundary, and at this point, R_λ is minimized, equivalent to the offline SVDD anomaly detection model. The choice of the detection threshold determines the model's tolerance to anomalous samples in online anomaly detection. A lower detection threshold leads to a larger detection boundary relative to the decision boundary, resulting in a higher misclassification rate of anomalous samples. Through statistics, it is observed that during the actual operation of bearings, anomalous samples are generally located outside the boundary at twice the radius from the center of the bearing's hypersphere. Therefore, setting the detection boundary threshold λ to 0.5 and considering this boundary as the detection boundary will enhance the accuracy of anomaly detection.

The establishment of the detection boundary will address the challenge of misclassifying anomalous samples during the updating process. It will enable the model to achieve precise identification of anomalous samples during the updating process, thereby improving the accuracy of online bearing anomaly detection.

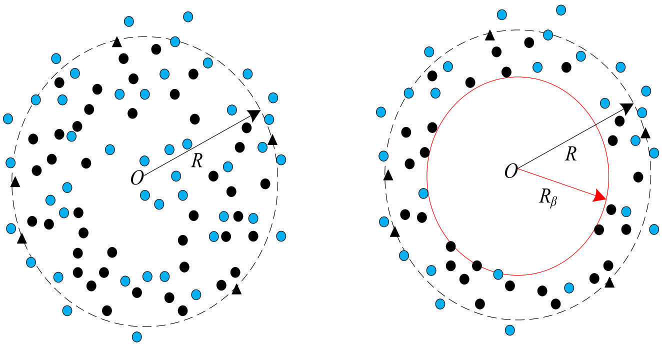

Figure 2. Illustration of elimination boundary.

2.3. A dual-screening mechanism update strategy for support vector

The problem of misclassification of samples during the model updating process has been resolved. However, the improvement of model update speed remains a key research focus in current online anomaly detection.

2.3.1. Preliminary screening mechanism based on the Elimination Boundary. To enhance the speed of model updates, a new challenge arises in the sample selection during the model updating process for ISVDD. For online anomaly detection, the selection of samples participating in updates can be categorized into three types: using all old samples, using all new samples, and using a combination of new and old samples [22]. Using only old samples cannot capture the degradation trend of bearing equipment, making the model unable to adapt to new samples. SVDD is sensitive to noise, and using only new samples may lead to significant bias in the new model due to incidental reasons such as sensor acquisition errors and environmental interference [23]. Using only old samples and using only new samples both fail to adapt to the characteristics of model updates over the entire lifespan of bearing equipment, and using synthetic samples combining new and old data can effectively address this challenge. However, if all new and old samples are used to train SVDD, as the number of updates increases during the degradation of bearing equipment performance, the number of samples involved in training will also increase, leading to a progressively higher computational complexity. This makes it impossible to achieve timely monitoring of bearing status.

To address the challenge of improving model update speed, a dual-screening mechanism update strategy for supporting vectors is proposed. The first step involves a preliminary screening of samples within the hypersphere. Similar to the principle of establishing the detection boundary, the elimination boundary is established within the hypersphere decision boundary, as shown in figure 2. The black dashed line represents the original model decision boundary, the red solid line represents the elimination boundary, and on the left, new and old samples are added to the hypersphere original model, while on the right, the elimination boundary eliminates new and old samples added to the original model.

The establishment of the Elimination Boundary aims to conduct a preliminary screening for the participating nSVs in the update. Due to the degradation of bearing performance, under normal circumstances, samples on the SVDD model tend to diverge towards the surroundings. There may also be a slight overall offset towards the sides. Therefore, the samples becoming SVs for the next model do not necessarily all come from SVKs; they may also come from the original model's SVs and samples near the decision boundary inside. The establishment of the Elimination Boundary is used to retain samples near the decision boundary, providing fewer suitable nSVs for the model's update. By setting a suitable elimination threshold, an appropriate elimination boundary is established, leading to the elimination of all new and old samples falling within the elimination boundary. Setting the elimination threshold:

$$\beta = \frac{R_\beta}{R}, \ 0 < \beta < 1 \tag{8}$$

where R_β is the radius of the elimination boundary. When R_β is minimum zero, it means that no elimination is carried out, and when R_β is maximum R, it means that all the old and new samples within the decision boundary of the previous model are eliminated, and only SVKs are retained. Failure to eliminate samples will lead to an increasing number of samples participating in the update training. Eliminating all new and old samples within the decision boundary will result in no old samples participating in the update training. Therefore, the minimum value of R_β should be greater than zero, and the maximum should be less than R. The selection of the elimination threshold determines the model's punishment level for samples within the decision boundary of the online anomaly detection model. The higher the elimination threshold, the larger the elimination boundary relative to the decision boundary, resulting in more elimination of new and old samples, and consequently, fewer participating nSVs in the update. In the actual operation process of bearings, newly added samples generally disperse uniformly outward along the sphere center as the bearing performance degrades. However, considering the possibility of overall deviation, combined with the statistical results of multiple experiments, the elimination threshold β is set to 0.54 as the most suitable. By setting the current boundary as the elimination boundary, it is possible to significantly reduce the number of samples participating in the update while maintaining the robustness of the model update process.

2.3.2. In-depth screening mechanism based on time density weight. The second step of the dual-screening mechanism update strategy is to propose a screening mechanism based on time density weight on the basis of the preliminary screening at the elimination boundary. It performs in-depth screening on the remaining nSVs to involve fewer samples in the update, further enhancing the real-time capability of online anomaly detection in bearings.

In the process of model updating, new and old samples participate in the update training together, but their contributions to the model are different. The model at the current moment should be established based on the samples collected at the current moment to achieve anomaly detection at the current moment. Online anomaly detection for bearings is aimed at detecting anomalies at each moment during the entire lifecycle of the bearing's operation, so new samples are crucial for the establishment of a new model. In comparison, old samples only provide a past reference for the new model, allowing the new model to avoid significant deviations caused by noise interference. As the number of updates increases, the contribution of old samples to the new model will gradually decrease. Therefore, the time decay weight function is proposed for the sample x_i:

$$W_i(t) = \mathrm{e}^{-\alpha t}, \alpha > 0 \tag{9}$$

where t is the time step, $W_i(t)$ represents the time decay weight of the ith sample at time t. α is the decay coefficient, and by adjusting α, the decrease rate of the time decay weight can be flexibly controlled. The value of α can be empirically determined through multiple experiments. New samples are assigned the highest time decay weights, while old samples are assigned relatively lower time decay weights. With the increase in the number of updates, the time decay weight of each sample decreases sequentially. The concept of time decay weights for samples is relative between new and old samples. When new samples are input in the next time step, the previous new samples become old samples for the next time step. Therefore, the time decay weight of a sample itself only decreases sequentially, approaching zero gradually, but never reaching zero. The lower the time decay weight of a sample, the earlier it participated in model updating training, the more times it participated in updates, the smaller its contribution to the new model, and the less likely it is to become the SVs.

The time decay weights of samples provides temporal relative weights, but for the selection of nSVs, the temporal weights of samples cannot ignore the spatial density information. In the SVDD model updating process, the influence between samples is continuous, and this influence is not only temporal but also spatial, where neighboring samples will inevitably affect the sample at that point in space. Therefore, based on the time decay weight, the idea of K-nearest neighbors (KNN) can be borrowed [24], proposing the time density weight function for each the sample x_i:

$$W_{\mathrm{TD}_i} = \frac{\sum_{i=1}^{k} W_i(t)}{k} \tag{10}$$

where $W_i(t)$ is the time decay weight of the sample x_i, k is the number of nearest neighbors for the sample x_i, and the value of k can be determined through multiple experiments based on experience. The time density weight function calculates the average of the time decay weights of the k nearest samples to the sample x_i, excluding the sample x_i itself. This ensures that the sample focuses more on the influence of nearby samples on itself, avoiding noise interference that may cause some samples to deviate from the overall impact, thereby ensuring the robustness of the model.

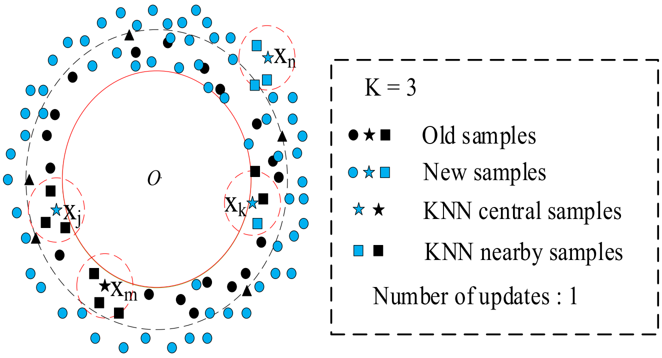

Figure 3. Details of samples distribution with time density weights.

The time density weight function considers the spatial relationship between samples on the basis of time decay weights, fully reflecting the evolution and correlation of samples in both time and space. The lower the time density weight of a sample, the lower the temporal decay weight in its vicinity. In spatial terms, this is manifested by higher isolation in time decay weights, such as newer samples with higher isolation or samples with time decay weights similar to their neighbors. Consequently, the likelihood of becoming SVs for that sample is lower. Higher isolation in time decay weights in space implies a higher probability of being affected by noise interference, resulting in a lower likelihood of becoming SVs naturally.

The detailed distribution chart of time density weights for samples is shown in figure 3. Taking the first model update as an example, the time decay weights for old samples is $W_i(1) = \mathrm{e}^{-\alpha}$, and for new samples, the time decay weights are $W_i(0) = 1$.Therefore, the time density weight for the sample x_j is $W_{\mathrm{TD}_j} = \mathrm{e}^{-\alpha}$, for the sample x_m is $W_{\mathrm{TD}_m} = \mathrm{e}^{-\alpha}$, for the sample x_k is $W_{\mathrm{TD}_k} = \frac{2\mathrm{e}^{-\alpha}+1}{3}$, and for the sample x_n is $W_{\mathrm{TD}_n} = 1$, with $W_{\mathrm{TD}_j} = W_{\mathrm{TD}_m} < W_{\mathrm{TD}_k} < W_{\mathrm{TD}_n}$.The sample x_n has the highest time density weight, making it the most likely to become SVs. The sample x_j and x_m have equal and lowest time density weights, making them the least likely to become SVs after this round of updates and are likely to be eliminated. In particular, the sample x_j, compared to the sample x_k, is a new sample with the highest isolation in time decay weight in space, making it the most likely to be affected by noise and should be excluded from becoming SVs. It should be eliminated in this round of in-depth screening mechanism.

By eliminating a batch of samples with the lowest time density weights and making the remaining samples participate in the updating training, the in-depth screening after the preliminary screening of nSVs will be realized, and the samples participating in the updating will be further reduced.

The proposal of the dual-screening mechanism update strategy involves layer-by-layer screening of nSVs, effectively reducing the number of samples involved in the update while maintaining model robustness. This results in a significant reduction in computational workload, thereby enhancing the speed of model updates and improving the real-time capability of online bearing anomaly detection.

2.4. Online bearing anomaly detection framework based on time density weight

During the whole life cycle of bearings, various factors such as harsh working environment, external interference and performance degradation may lead to abnormal vibration signals collected, so the monitoring of vibration data is of great significance for the online abnormal detection of bearings [25].The health stage of a bearing constitutes 80%–90% of its entire lifespan [2, 26]. During the health stage, the vibration data generally exhibits a long-term, small, and stable fluctuation trend. Subsequently, it enters a short-term degradation stage, where the amplitude of the bearing vibration data gradually increases. Finally, it reaches an extremely short failure stage, marked by a sudden and significant increase in the amplitude of vibration data. Entering this stage signifies reaching the end of the operational lifespan, indicating that the bearing is no longer operational. During the entire lifespan of the bearing, the vibration data fluctuates around zero, with both positive and negative amplitudes roughly balanced. Additionally, a small amount of anomalous data is generated throughout the entire lifespan stage.

Based on the characteristics of the vibration data throughout the entire lifespan of the bearing and to meet the requirement of improving real-time performance under limited resources, we propose a TISVDD online anomaly detection framework. This framework integrates the detection boundary and elimination boundary into a dual-boundary SVDD model. Subsequently, samples are assigned time density weights for screening, allowing the model to dynamically update with the changing data in the time dimension. This approach aims to achieve robust online anomaly detection for bearings.

The vibration signals of the bearing are continuously acquired online as a batch to form the online degradation dataset. The initial set of samples is used to train the dual-boundary SVDD model to make it complete. New samples are then introduced to determine if they satisfy the KKT conditions. If all new samples satisfy the KKT conditions, the model does not undergo an update, and the evaluation is deferred until the next batch of samples arrives. In case there are new samples violating the KKT conditions, the detection boundary is employed to identify abnormal samples among them. The remaining new samples are combined with the old samples from the previous stage. Preliminary screening of nSVs is carried out using the Elimination Boundary. Time density weights are assigned to samples, and in-depth screening of nSVs is performed based on these weights. Finally, the updated dual-boundary SVDD model is constructed using the remaining nSVs. This process is repeated until the bearing enters the fault stage, at which point the model ceases to update, and the bearing becomes inoperative. The overall framework of TISVDD for online anomaly detection is illustrated in figure 4, and the algorithmic flowchart of TISVDD is shown in figure 5.

The proposal of the TISVDD online anomaly detection framework based on bearings starts from the characteristics of bearing vibration data. This allows the model to closely follow data changes for stable updates, effectively defining and removing abnormal samples during updates. Layered screening of nSVs significantly reduces the number of samples involved in updates, achieving an effective improvement in update speed and enhancing the accuracy and real-time performance of online bearing anomaly detection.

3. Simulation

3.1. Data setting

To verify the effectiveness and superiority of the proposed method in this experiment, we utilize the publicly available XJTU-SY Bearing Datasets to validate the first two innovations. This dataset is gathered from the bearing acceleration degradation test platform of Xi'an Jiaotong University's joint laboratory. It aims to demonstrate the enhancement in anomaly detection accuracy during the model update process, as proposed in the first innovation, and to confirm the improvement in model update speed during the process, as proposed in the second innovation. Subsequently, the validity of the online anomaly detection of the TISVDD model, proposed as the third innovation, is verified using a real-time dataset collected on the online hardware platform. The online hardware platform can real-time capture bearing vibration data and perform online detection, as illustrated in figure 6.

The XJTU-SY bearing dataset has a sampling frequency of 25.6 kHz, recording 32 768 samples of lateral and longitudinal vibration data in each sampling. This experiment studied the vibration data of three different bearings under conditions of 2100 RPM speed, 35 Hz frequency, and 12 kN load, ranging from normal to failure states. Experimental verification shows that training the SVDD model with 500 samples makes the model complete. Therefore, the degradation dataset for each file includes the first 500 samples for subsequent experiments, as shown in table 1.

3.2. Validation of TISVDD classification strategy

In this section, the effectiveness of the classification strategy proposed in this research to prevent misclassification of samples during the updating process is verified. It aims to demonstrate the improvement in online anomaly detection accuracy compared to traditional methods and current mainstream methods. Three metrics are selected for evaluation: average accuracy (AA), and F-score. Recall (TPR) is the score for true positives (TP), specificity (TNR) is the score for true negatives (TN), and precision (TPP) provides a balance between TP and FP accuracy. F-score is a highly suitable method for evaluating the accuracy of anomaly detection in imbalanced datasets. Its value is the harmonic mean of TPR and TPP, and it becomes significant only when both TPR and TPP are high. The formula is as follows:

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{11}$$

$$\mathrm{TNR} = \frac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FP}} \tag{12}$$

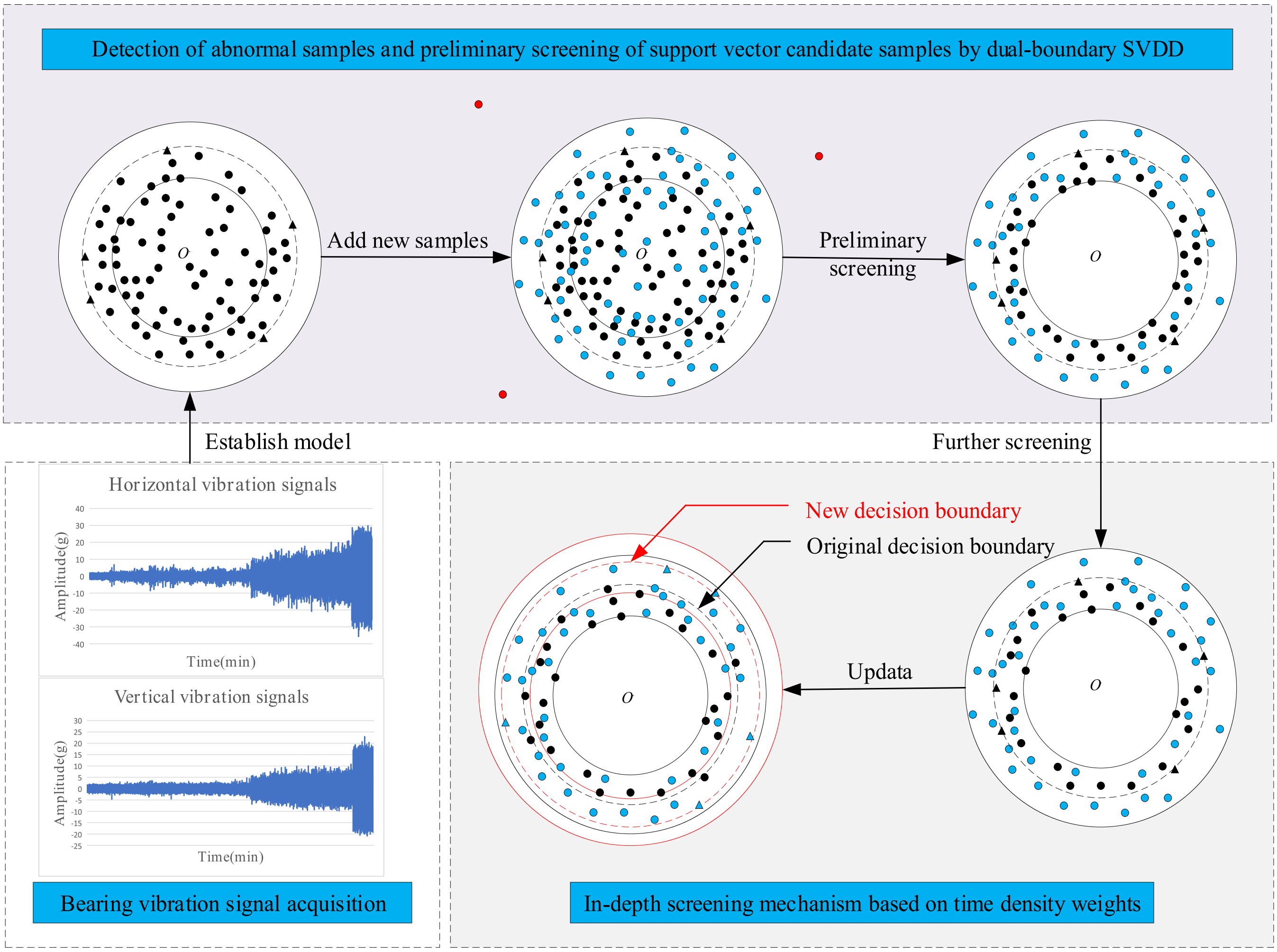

Figure 4. Overall framework of TISVDD online anomaly detection based on bearings.

$$\mathrm{TPP} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{13}$$

$$F-\mathrm{score} = \frac{2*\mathrm{TPR}*\mathrm{TPP}}{\mathrm{TPR}+\mathrm{TPP}}. \tag{14}$$

The TISVDD algorithm proposed in this research is compared with standard ISVDD, KISVDD, NISVDD, and HISVDD algorithms on three bearing datasets for online anomaly detection accuracy using the average F-score evaluation metric. The results are shown in table 2.

According to table 2, the TISVDD algorithm proposed in this research has the highest F-score on all three bearing datasets compared to other methods. This indicates that the performance of TISVDD in terms of online anomaly detection accuracy is the best. It is evident that the classification strategy of the TISVDD algorithm accurately distinguishes between abnormal samples and degraded normal samples in SVKs, effectively identifying abnormal samples during the update process. Moreover, it achieves an improvement in online anomaly detection accuracy compared to other algorithms.

To visually illustrate the adaptability of the TISVDD algorithm in online anomaly detection to different sample quantities during the update process, we conducted a comparative bar chart of the classification accuracy of the proposed TISVDD with the current mainstream algorithm HISVDD and the traditional incremental algorithm standard ISVDD across varying numbers of incremental samples. The results are shown in figure 7.

From figure 7, it is visually evident that TISVDD, proposed in this research, exhibits lower classification accuracy in online anomaly detection when there are fewer samples, indicating that the model is not yet complete. Similar to the other two methods, with too few samples, the differences in features between anomaly and normal samples are not as distinct, making the model prone to false positives or missing positives. As the number of added samples increases, the model gradually becomes complete, and the classification accuracy of TISVDD stabilizes above 96%, significantly higher than the other two methods. This suggests a notable online anomaly detection performance, demonstrating good adaptability to varying sample quantities. Therefore, table 2 and figure 7 together validate the outstanding online anomaly detection

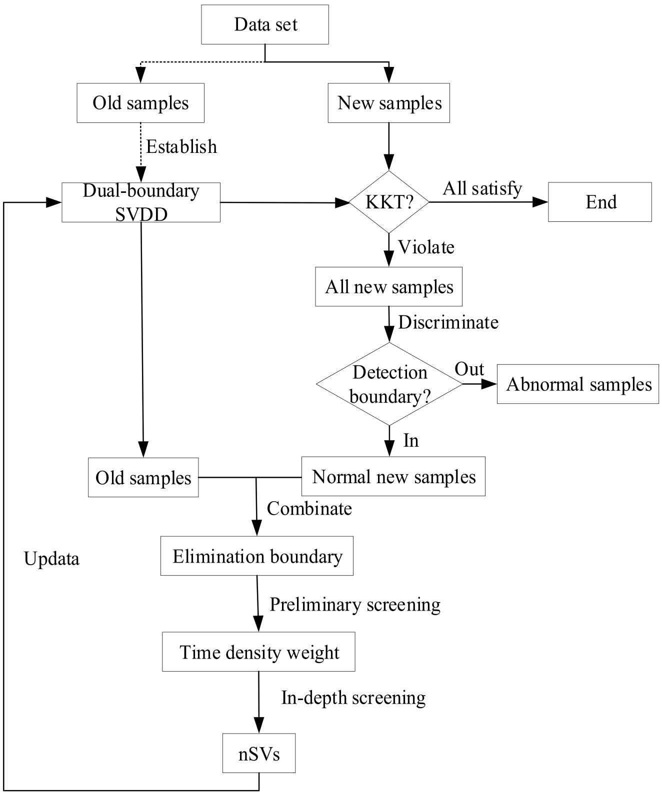

Figure 5. Algorithm flowchart of TISVDD.

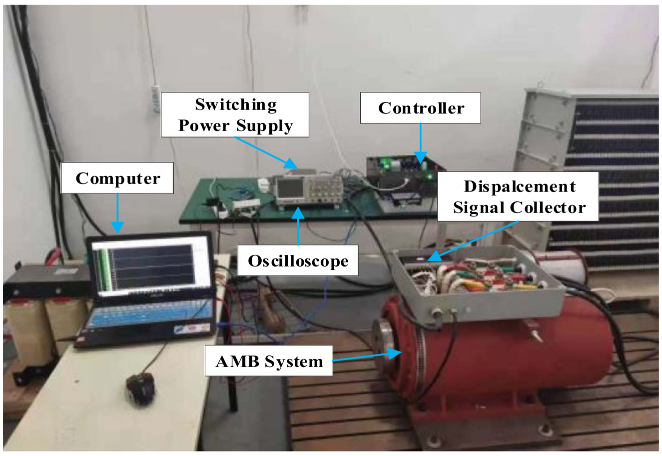

Figure 6. Online hardware platform.

Table 1. Degradation data set of bearings in different failure states.

Bearing dataset	Number of files	Fault element	Number of samples per file
Bearing 1_1	123	Outer race	500
Bearing 1_4	122	Cage	500
Bearing 1_5	52	Inner and outer race	500

capabilities of TISVDD and its improvement in online anomaly detection accuracy compared to other online anomaly detection methods.

Table 2. F-score on data sets.

Bearing data set	ISVDD	KISVDD	NISVDD	HISVDD	TISVDD
Bearing 1_1	86.55	90.25	94.26	94.46	97.81
Bearing 1_4	87.36	92.33	93.58	92.68	97.58
Bearing 1_5	85.96	91.28	93.85	93.57	96.38

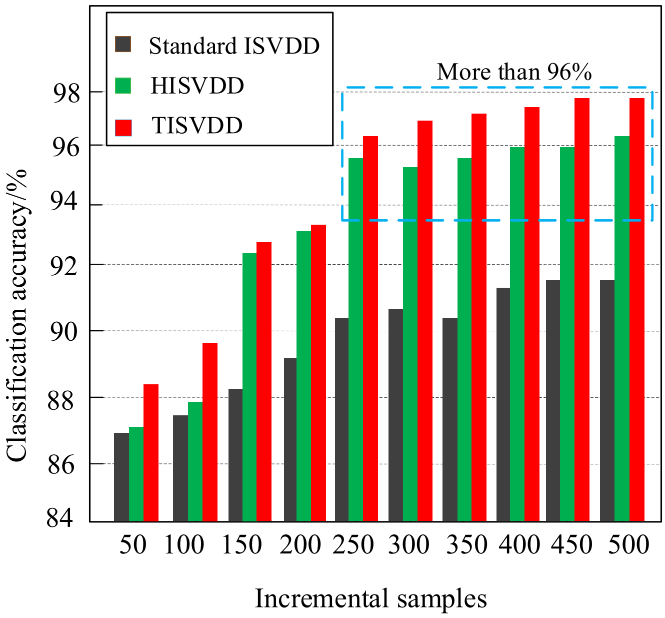

Figure 7. Classification accuracy of TISVDD.

3.3. Validation of TISVDD update strategy

To demonstrate the TISVDD algorithm's ability to maintain good robustness during model updates and its improvement in update speed, we conducted a comparison with several mainstream algorithms. We refer to the model trained on the newly added dataset at each moment as the true model at that moment, the model trained by standard ISVDD. The TISVDD algorithm proposed in this research and the models updated by other mainstream algorithms serve as predictive models. The similarity between the predictive models and the true models is used as the metric for comparing our method with other algorithms. During the model update process, only the hypersphere center c, hypersphere radius R, and support vectors SVs may undergo changes. Since c and R can be obtained from SVs, this allows us to calculate the Spearman correlation coefficient of SVs between the predictive models generated by our algorithm and other algorithms and the true models, serving as a measure of similarity.

In the online update algorithm, the SVs of the true model serve as the dataset rSVs, and the SVs of the predicted model serve as the dataset pSVs. Assuming that the data lengths of rSVs and pSVs are both n, where any element value i in the set is represented as $ri, pi\,(1 \leqslant i \leqslant n)$, perform simultaneous ascending or descending sorting on all elements in the data rSVs and pSVs. This results in newly sorted sets RSVs and PSVs, where the element Ri is the ith element in RSVs, and Pi is the ith element in PSVs. Subtract the corresponding position

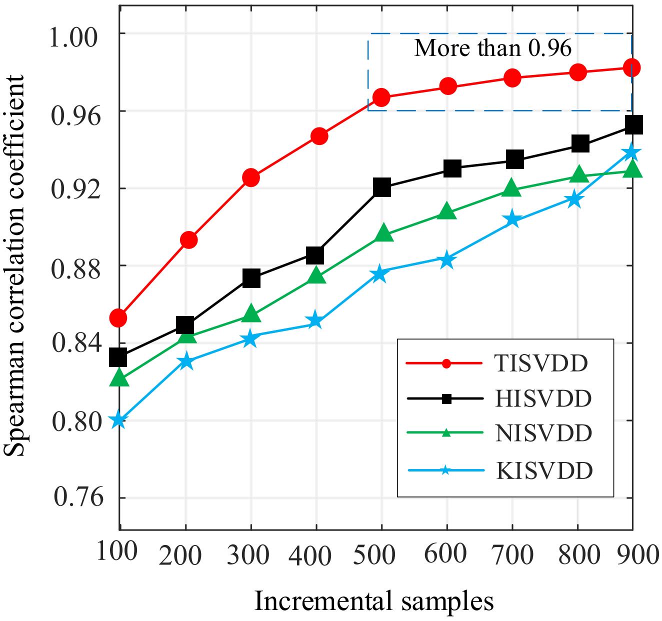

Figure 8. Average spearman correlation coefficients for different algorithms.

elements in the sets RSVs and PSVs to obtain a rank difference set D, where $di = ri - pi, di \in D, 1 \leqslant i \leqslant n$. With m being the number of updates in the degradation stage, the formula for calculating the Spearman rank correlation coefficient between the support vectors rSVs and pSVs for the jth update is as follows:

$$\rho_j = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n\left(n^2 - 1\right)}. \tag{15}$$

The average Spearman rank correlation coefficient throughout the entire lifespan update process is calculated as follows:

$$\rho = \frac{\sum_{j=1}^{m} \rho_j}{m}. \tag{16}$$

The TISVDD algorithm proposed in this research, along with the standard ISVDD, KISVDD, NISVDD, and HISVDD algorithms, undergo online updates on the aforementioned bearing degradation datasets. The average Spearman rank correlation coefficients are shown in figure 8, and the average update time for individual models is presented in table 3.

From figure 8, it can be seen that the TISVDD algorithm, proposed in this research, exhibits the highest average Spearman rank correlation coefficient during the model updating process across all ranges of Incremental samples. This indicates that the model updated by the TISVDD algorithm has the highest similarity with the true model. In cases with fewer samples, the correlation coefficient is lower because the model is not complete. As the number of samples increases, the model gradually becomes complete. The correlation coefficient of the TISVDD algorithm's updated model stabilizes above 0.96, demonstrating that the update strategy of the TISVDD algorithm is well-adapted to model updates caused by the degradation of bearing performance. It can adapt to changes in Incremental samples and exhibits good robustness. Additionally, from table 3, it can be seen that the TISVDD algorithm has lower average update times for individual models significantly across the three bearing datasets compared to other algorithms. This confirms that the TISVDD algorithm effectively reduces computational workload and improves update speed. Considering figure 8 and table 3 together, it can be concluded that the TISVDD algorithm maintains model robustness during the update process while achieving an improvement in model update speed, enhancing the real-time capability of bearing online anomaly detection.

Table 3. The data sets used in the algorithm accuracy tests.

	Average update time (s)				
Bearing dataset	ISVDD	KISVDD	NISVDD	HISVDD	TISVDD
Bearing 1_1	25.18	1.54	1.44	1.24	0.82
Bearing 1_4	22.35	1.62	1.38	1.21	0.79
Bearing 1_5	20.16	1.49	1.31	1.15	0.72

3.4. *Validation of TISVDD model updating*

For validating the effectiveness of the TISVDD algorithm in model updating proposed in this research, we utilized an online hardware platform to collect real-time bearing vibration data. The process diagram of updating the model using the TISVDD algorithm is presented in MATLAB, as shown in figure 9.

From figure 9, the entire process of TISVDD algorithm updating the model can be intuitively observed. In figure 9(b), compared to figure 9(a), the dual-boundary SVDD is used for training, and the elimination boundary screens the old samples of the original model. The number of samples is reduced by 61.6%, but the model hardly changes. It can be seen that the establishment of the elimination boundary of the dual-boundary SVDD allows the model to maintain robustness while significantly reducing the number of samples, effectively reducing computation. Additionally, in figure 9(d), a depth filtering mechanism based on time density weight is applied on top of figure 9(c) to train the model for complete updates, reducing the number of samples by 41.5%. For the entire process of TISVDD algorithm updating the model, considering the participation of both new and old samples in updates results in a reduction of 71.8% in samples, while considering either new or old samples alone reduces the samples by 43.6%, achieving a significant reduction in samples and a substantial decrease in computation, thereby enhancing update speed. From figures 9(a) to (d), the TISVDD algorithm stably updates the original model to the new model

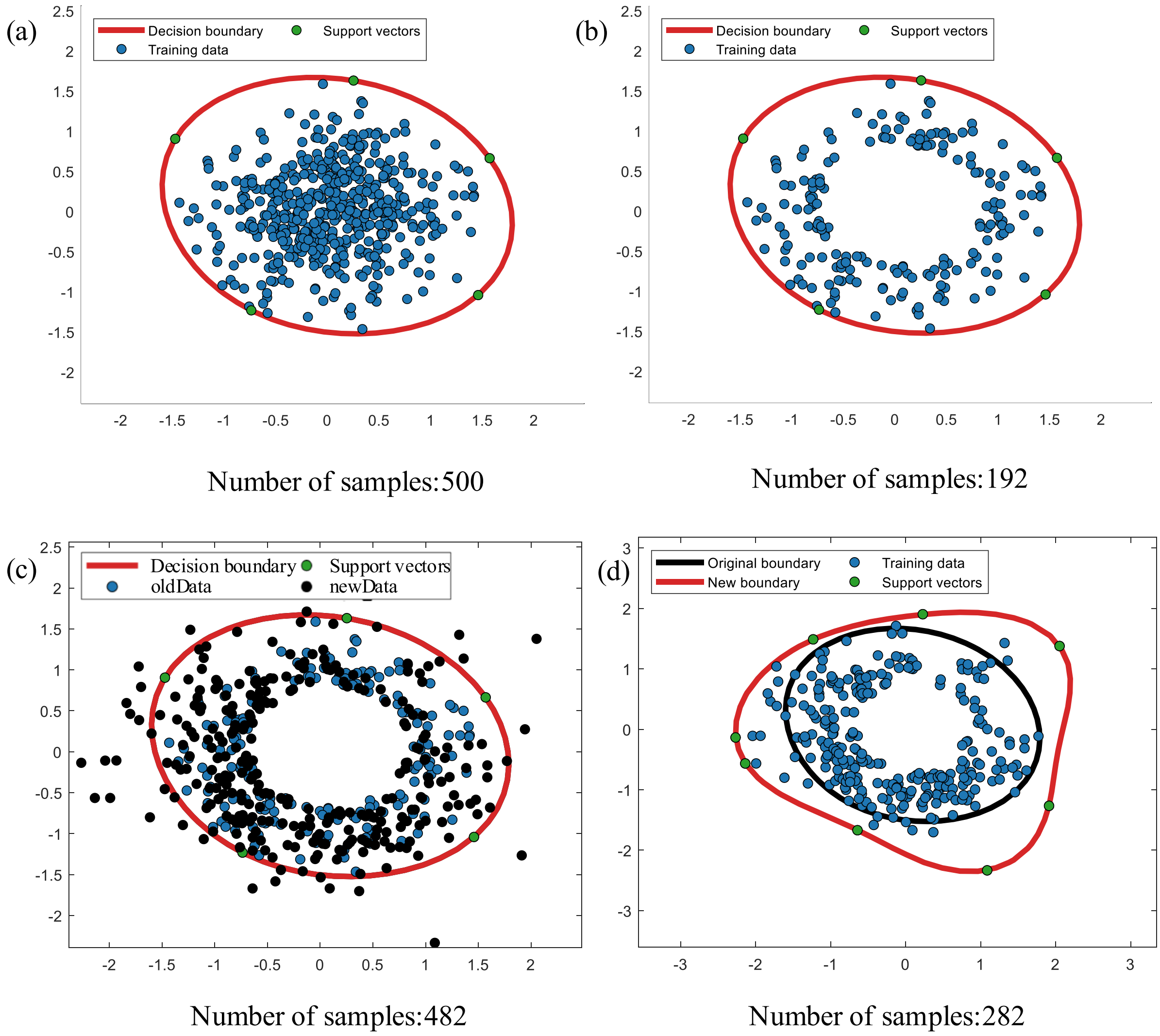

Figure 9. TISVDD model updates: (a) SVDD training; (b) dual-boundary SVDD training; (c) preliminary screening; (d) in-depth screening.

following the data changes, fully demonstrating the effectiveness of TISVDD algorithm in updating online anomaly detection models and achieving robust online anomaly detection for bearings.

4. Conclusion

This manuscript proposes an online anomaly detection model, TISVDD, designed for the online operational scenarios of bearings. The model innovatively achieves online anomaly detection for bearings. TISVDD is designed to improve the accuracy and update speed of online anomaly detection for bearing operational data in current ISVDD methods. Thorough validation is conducted using the XJTU-SY bearing dataset and real-time data collected from an online hardware platform. We confirm the rationality and effectiveness of the proposed method and obtain a series of valuable conclusions, providing important insights for future research. Firstly, we introduce a classification strategy to prevent samples misclassification during the update process, establish the Detection Boundary, significantly enhance the model's ability to identify abnormal samples during the update process, prevent misclassification of abnormal samples, highlighting the uniqueness of our method. Secondly, we propose a dual-screening mechanism update strategy for SVs. After preliminary screening of new and old samples that may become SVs based on the establishment of the Elimination Boundary, an in-depth screening is performed on the remaining samples based on time density weight, thereby greatly reducing the number of samples involved in the update, effectively reducing the computational load, and enhancing the real-time performance of bearing online anomaly detection. This demonstrates the novelty of our method. Finally, we propose an ISVDD online anomaly detection framework based on time density weight, merging the detection boundary and elimination boundary into a dual-boundary SVDD model. Subsequently, samples are assigned time density weight for screening, allowing the model to dynamically update with changes in bearing vibration data in the time dimension, achieving robust online anomaly detection for bearings. In conclusion, the proposed method is not only theoretically innovative but also exhibits significant novelty in practical applications. It provides profound insights for further research in the field of online anomaly detection for bearings.

However, this method still has some shortcomings. For example, the updating strategy proposed in this paper still requires retaining a large number of samples for training updates. If too many samples are reduced through this method, the model after updating training will differ significantly from the model trained using all samples. In the future, it may be beneficial to optimize this updating strategy to involve fewer samples in the update training process, greatly improving the update speed. This would enable more real-time online anomaly detection when deployed on resource-limited embedded devices.

Data availability statement

All data that support the findings of this study are include within the article (and any supplementary files).

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